

SUMMARY

There are two types of options: calls and puts. A call option gives the holder the right to buy the underlying asset for a certain price by a certain date. A put option gives the holder the right to sell the underlying asset by a certain date for a certain price. There are four possible positions in options markets: a long position in a call, a short position in a call, a long position in a put, and a short position in a put. Taking a short position in an option is known as writing it. Options are currently traded on stocks, stock indices, foreign currencies, futures contracts, and other assets.

An exchange must specify the terms of the option contracts it trades. In particular, it must specify the size of the contract, the precise expiration time, and the strike price. In the United States one stock option contract gives the holder the right to buy or sell 100 shares. The standard expiration of a stock option contract is the third Friday of the expiration month. Options with several different expiration months trade at any given time. Strike prices are at $\$2\frac{1}{2}$, $\$5$, or $\$10$ intervals, depending on the stock price.

The terms of a stock option are not normally adjusted for cash dividends. However, they are adjusted for stock dividends, stock splits, and rights issues. The aim of the adjustment is to keep the positions of both the writer and the buyer of a contract unchanged.

Most option exchanges use market makers. A market maker is an individual who is prepared to quote both a bid price (at which he or she is prepared to buy) and an ask price (at which he or she is prepared to sell). Market makers improve the liquidity of the market and ensure that there is never any delay in executing market orders. They themselves make a profit from the difference between their bid and ask prices (known as their bid–ask spread). The exchange has rules specifying upper limits for the bid–ask spread.

Writers of options have potential liabilities and are required to maintain a margin account with their brokers. If it is not a member of the Options Clearing Corporation, the broker will maintain a margin account with a firm that is a member. This firm will in turn maintain a margin account with the Options Clearing Corporation. The Options Clearing Corporation is responsible for keeping a record of all outstanding contracts, handling exercise orders, and so on.

Not all options are traded on exchanges. Many options are traded in the over-the-counter (OTC) market. An advantage of over-the-counter options is that they can be tailored by a financial institution to meet the particular needs of a corporate treasurer or fund manager.

FURTHER READING

Chicago Board Options Exchange. *Margin Manual*. Available online at the CBOE website: www.cboe.com.

Short Concept Questions

- 10.1. What is the difference between an American and a European option?
- 10.2. What are the four types of option positions?
- 10.3. Give examples of three index options that trade on the CBOE?

- 10.4. What is a flex option?
- 10.5. Which of the following lead to adjustments to the terms of an exchange-traded option: (a) stock splits, (b) stock dividends, (c) cash dividends?
- 10.6. What is a position limit? What is its purpose?
- 10.7. What is the potential advantage to a corporation of trading an over-the-counter option rather than an exchange-traded option?
- 10.8. Explain why margin accounts are required when clients write options but not when they buy options.
- 10.9. A stock option is on a February, May, August, and November cycle. What options trade on (a) April 1 and (b) May 30?
- 10.10. Explain carefully the difference between writing a put option and buying a call option.

Practice Questions

- 10.11. An investor buys a European put on a share for \$3. The stock price is \$42 and the strike price is \$40. Under what circumstances does the investor make a profit? Under what circumstances will the option be exercised? Draw a diagram showing the variation of the investor's profit with the stock price at the maturity of the option.
- 10.12. An investor sells a European call on a share for \$4. The stock price is \$47 and the strike price is \$50. Under what circumstances does the investor make a profit? Under what circumstances will the option be exercised? Draw a diagram showing the variation of the investor's profit with the stock price at the maturity of the option.
- 10.13. An investor sells a European call option with strike price of K and maturity T and buys a put with the same strike price and maturity. Describe the investor's position.
- 10.14. A company declares a 2-for-1 stock split. Explain how the terms change for a call option with a strike price of \$60.
- 10.15. "Employee stock options issued by a company are different from regular exchange-traded call options on the company's stock because they can affect the capital structure of the company." Explain this statement.
- 10.16. Suppose that a European call option to buy a share for \$100.00 costs \$5.00 and is held until maturity. Under what circumstances will the holder of the option make a profit? Under what circumstances will the option be exercised? Draw a diagram illustrating how the profit from a long position in the option depends on the stock price at maturity of the option.
- 10.17. Suppose that a European put option to sell a share for \$60 costs \$8 and is held until maturity. Under what circumstances will the seller of the option (the party with the short position) make a profit? Under what circumstances will the option be exercised? Draw a diagram illustrating how the profit from a short position in the option depends on the stock price at maturity of the option.
- 10.18. Consider the following portfolio: a newly entered-into long forward contract on an asset and a long position in a European put option on the asset with the same maturity as the forward contract and a strike price that is equal to the forward price of the asset at the time the portfolio is set up. Show that it has the same value as a European call option

- with the same strike price and maturity as the European put option. Deduce that a European put option has the same value as a European call option with the same strike price and maturity when the strike price for both options is the forward price.
- 10.19. A trader buys a call option with a strike price of \$45 and a put option with a strike price of \$40. Both options have the same maturity. The call costs \$3 and the put costs \$4. Draw a diagram showing the variation of the trader's profit with the asset price.
- 10.20. Explain why an American option is always worth at least as much as a European option on the same asset with the same strike price and exercise date.
- 10.21. Explain why an American option is always worth at least as much as its intrinsic value.
- 10.22. The treasurer of a corporation is trying to choose between options and forward contracts to hedge the corporation's foreign exchange risk. Discuss the advantages and disadvantages of each.
- 10.23. Consider an exchange-traded call option contract to buy 500 shares with a strike price of \$40 and maturity in 4 months. Explain how the terms of the option contract change when there is: (a) a 10% stock dividend; (b) a 10% cash dividend; and (c) a 4-for-1 stock split.
- 10.24. "If most of the call options on a stock are in the money, it is likely that the stock price has risen rapidly in the last few months." Discuss this statement.
- 10.25. What is the effect of an unexpected cash dividend on (a) a call option price and (b) a put option price?
- 10.26. Options on a stock are on a March, June, September, and December cycle. What options trade on (a) March 1, (b) June 30, and (c) August 5?
- 10.27. Explain why the market maker's bid-ask spread represents a real cost to options investors.
- 10.28. A U.S. investor writes five naked call option contracts. The option price is \$3.50, the strike price is \$60.00, and the stock price is \$57.00. What is the initial margin requirement?
- 10.29. Calculate the intrinsic value and time value from the mid market (average of bid and ask) prices for the September call options in Table 1.2. Do the same for the September put options in Table 1.3. Assume in each case that the current mid market stock price is \$316.00.
- 10.30. A trader writes 5 naked put option contracts with each contract being on 100 shares. The option price is \$10, the time to maturity is 6 months, and the strike price is \$64.
- (a) What is the margin requirement if the stock price is \$58?
 - (b) How would the answer to (a) change if the rules for index options applied?
 - (c) How would the answer to (a) change if the stock price were \$70?
 - (d) How would the answer to (a) change if the trader is buying instead of selling the options?
- 10.31. "If a company does not do better than its competitors but the stock market goes up, executives do very well from their stock options. This makes no sense." Discuss this viewpoint. Can you think of alternatives to the usual employee stock option plan that take the viewpoint into account.
- 10.32. On July 20, 2004, Microsoft surprised the market by announcing a \$3 dividend. The ex-dividend date was November 17, 2004, and the payment date was December 2, 2004. Its

stock price at the time was about \$28. It also changed the terms of its employee stock options so that each exercise price was adjusted downward to

$$\text{Predividend exercise price} \times \frac{\text{Closing price} - \$3.00}{\text{Closing price}}$$

The number of shares covered by each stock option outstanding was adjusted upward to

$$\text{Number of shares predividend} \times \frac{\text{Closing price}}{\text{Closing price} - \$3.00}$$

“Closing Price” means the official NASDAQ closing price of a share of Microsoft common stock on the last trading day before the ex-dividend date. Evaluate this adjustment.

Short Concept Questions

- 11.1. List six factors that affect stock option prices.
- 11.2. What is the lower bound for a European call option on a stock paying no dividends?
- 11.3. What is the lower bound for a European put option on a stock paying no dividends?
- 11.4. What is the put–call parity equation for a stock paying no dividends?
- 11.5. Under what circumstances should an American call option on a stock paying no dividends be exercised early?
- 11.6. Under what circumstances should an American put option on a stock paying no dividends be exercised early?

Practice Questions

- 11.7. What is a lower bound for the price of a 4-month call option on a non-dividend-paying stock when the stock price is \$28, the strike price is \$25, and the risk-free interest rate is 8% per annum?
- 11.8. What is a lower bound for the price of a 1-month European put option on a non-dividend-paying stock when the stock price is \$12, the strike price is \$15, and the risk-free interest rate is 6% per annum?
- 11.9. Give two reasons why the early exercise of an American call option on a non-dividend-paying stock is not optimal. The first reason should involve the time value of money. The second should apply even if interest rates are zero.
- 11.10. “The early exercise of an American put is a trade-off between the time value of money and the insurance value of a put.” Explain this statement.
- 11.11. Why is an American call option on a dividend-paying stock always worth at least as much as its intrinsic value. Is the same true of a European call option? Explain your answer.
- 11.12. The price of a non-dividend-paying stock is \$19 and the price of a 3-month European call option on the stock with a strike price of \$20 is \$1. The risk-free rate is 4% per annum. What is the price of a 3-month European put option with a strike price of \$20?
- 11.13. Explain why the arguments leading to put–call parity for European options cannot be used to give a similar result for American options.
- 11.14. What is a lower bound for the price of a 6-month call option on a non-dividend-paying stock when the stock price is \$80, the strike price is \$75, and the risk-free interest rate is 10% per annum?
- 11.15. What is a lower bound for the price of a 2-month European put option on a non-dividend-paying stock when the stock price is \$58, the strike price is \$65, and the risk-free interest rate is 5% per annum?
- 11.16. A 4-month European call option on a dividend-paying stock is currently selling for \$5. The stock price is \$64, the strike price is \$60, and a dividend of \$0.80 is expected in 1 month. The risk-free interest rate is 12% per annum for all maturities. What opportunities are there for an arbitrageur?

- 11.17. A 1-month European put option on a non-dividend-paying stock is currently selling for \$2.50. The stock price is \$47, the strike price is \$50, and the risk-free interest rate is 6% per annum. What opportunities are there for an arbitrageur?
- 11.18. Give an intuitive explanation of why the early exercise of an American put becomes more attractive as the risk-free rate increases and volatility decreases.
- 11.19. The price of a European call that expires in 6 months and has a strike price of \$30 is \$2. The underlying stock price is \$29, and a dividend of \$0.50 is expected in 2 months and again in 5 months. Risk-free interest rates (all maturities) are 10%. What is the price of a European put option that expires in 6 months and has a strike price of \$30?
- 11.20. Explain the arbitrage opportunities in Problem 11.19 if the European put price is \$3.
- 11.21. The price of an American call on a non-dividend-paying stock is \$4. The stock price is \$31, the strike price is \$30, and the expiration date is in 3 months. The risk-free interest rate is 8%. Derive upper and lower bounds for the price of an American put on the same stock with the same strike price and expiration date.
- 11.22. Explain carefully the arbitrage opportunities in Problem 11.21 if the American put price is greater than the calculated upper bound.
- 11.23. Prove the result in equation (11.7). (*Hint:* For the first part of the relationship, consider (a) a portfolio consisting of a European call plus an amount of cash equal to K , and (b) a portfolio consisting of an American put option plus one share.)
- 11.24. Prove the result in equation (11.11). (*Hint:* For the first part of the relationship, consider (a) a portfolio consisting of a European call plus an amount of cash equal to $D + K$, and (b) a portfolio consisting of an American put option plus one share.)
- 11.25. Consider a 5-year call option on a non-dividend-paying stock granted to employees. The option can be exercised at any time after the end of the first year. Unlike a regular exchange-traded call option, the employee stock option cannot be sold. What is the likely impact of this restriction on the early-exercise decision?
- 11.26. Use the software DerivaGem to verify that Figures 11.1 and 11.2 are correct.
- 11.27. What is the impact (if any) of negative interest rates on:
 - (a) The put–call parity result for European options
 - (b) The result that American call options on non-dividend-paying stocks should never be exercised early
 - (c) The result that American put options on non-dividend-paying stocks should sometimes be exercised early.
 Assume that holding cash earning zero interest is not possible.
- 11.28. Calls were traded on exchanges before puts. During the period of time when calls were traded but puts were not traded, how would you create a European put option on a non-dividend-paying stock synthetically?
- 11.29. The prices of European call and put options on a non-dividend-paying stock with an expiration date in 12 months and a strike price of \$120 are \$20 and \$5, respectively. The current stock price is \$130. What is the implied risk-free rate?
- 11.30. A European call option and put option on a stock both have a strike price of \$20 and an expiration date in 3 months. Both sell for \$3. The risk-free interest rate is 10% per annum, the current stock price is \$19, and a \$1 dividend is expected in 1 month. Identify the arbitrage opportunity open to a trader.

- 11.31. Suppose that c_1 , c_2 , and c_3 are the prices of European call options with strike prices K_1 , K_2 , and K_3 , respectively, where $K_3 > K_2 > K_1$ and $K_3 - K_2 = K_2 - K_1$. All options have the same maturity. Show that

$$c_2 \leq 0.5(c_1 + c_3)$$

(*Hint:* Consider a portfolio that is long one option with strike price K_1 , long one option with strike price K_3 , and short two options with strike price K_2 .)

- 11.32. What is the result corresponding to that in Problem 11.31 for European put options?